

HAMZA JAWAD

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[Portfolio](#) | [Github](#) | [LinkedIn](#) | [Kaggle](#)

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SUMMARY

Data Scientist / AI-ML Engineer with hands-on experience building end-to-end, production-style ML pipelines — feature stores, model registries, and live-deployed apps — across churn/LTV, air-quality, and crypto-volatility forecasting. Benchmarked 10 NLP architectures (RNN to BERT/T5) and built a live Kafka–Spark sentiment pipeline. Proficient in Python, Scikit-learn, TensorFlow, PySpark, SQL, MLflow, and Docker.

WORK EXPERIENCE

10Pearls — Data Scientist (Intern)

Jul 2026 – Sep 2026

- Built an **end-to-end MLOps pipeline** forecasting Lahore's Air Quality Index at **24, 48, and 72 hours**, improving forecast accuracy by **18%** over a naive persistence baseline.
- Trained and registered **LightGBM, XGBoost, and Prophet** models, cutting forecast error by 22% **RMSE** to clear the persistence-baseline gate on RMSE, MAE, and R^2 .
- Served forecasts through a **FastAPI backend and Streamlit dashboard**, containerized with Docker and automated via **GitHub Actions**, improving pipeline uptime by **65%**.

DecodeLabs — Data Scientist (Intern)

May 2026 – Jun 2026

- Built **4 end-to-end ML classification pipelines** across fraud detection, loan approval, attrition prediction, and fake job NLP using **XGBoost, LightGBM, and Scikit-learn** with SMOTE, improving minority-class recall by **27%**.
- Applied NLP classification on **17,880 job postings** using TF-IDF and threshold tuning; engineered features across datasets of **981 to 17,880 records**, achieving **88% F1-score**.
- Delivered business recommendations from all 4 projects — including fraud risk signals, HR retention strategies, and loan screening logic — backed by **5-fold cross-validated** results averaging **90% ROC-AUC**.

PROJECTS

Real-Time Social Media Sentiment Analysis

Python | Apache Kafka | PySpark | Spark Streaming

- Streamed **500 tweets/run** via **Apache Kafka** into PySpark Structured Streaming; trained a **4-class ML classifier** reaching 89% **accuracy**, improving classification accuracy by **14%** over the baseline.

Deep NLP: Sentiment Classification

Python | TensorFlow | BERT | DistilBERT | T5

- Benchmarked **10 NLP architectures** (RNN, LSTM, GRU, BiLSTM, BiGRU, TextCNN, Transformer Encoder, BERT, DistilBERT, T5) on a **balanced Amazon reviews dataset**; best model reached **90% F1-score**.

E-Commerce Churn & LTV Predictor

Python | Feast | MLflow | FastAPI | Streamlit

- Built a **two-model MLOps pipeline** for e-commerce churn and lifetime value on **50,000 rows** via a **Feast feature store** with CatBoost champions **improving holdout ROC-AUC by 16%** over the baseline.

Binance Crypto Volatility Predictor

Python | LightGBM | XGBoost | MLflow | FastAPI

- Built a local-first MLOps pipeline forecasting **24-hour-ahead volatility** for **4 crypto assets (BTC, ETH, BNB, SOL)** trained **LightGBM and XGBoost** models, cutting forecast error by **19%** vs a naive baseline.

EDUCATION

University of Central Punjab, Lahore

Oct 2023 – Present

Bachelor of Science in Data Science (CGPA : 3.77)

Punjab College, Lahore

Jun 2021 – Aug 2023

ICS, Physics

SKILLS AND CERTIFICATIONS

Technical Skills: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib), PySpark, R, SQL, Kafka, Tableau, Excel, ML, Deep Learning, NLP (TF-IDF, BERT), Feature Engineering, EDA, Gen AI, MLflow, DagsHub, n8n, Knime, Airflow, Docker

Interpersonal Skills: Data storytelling, analytical thinking, problem-solving, teamwork, communication, attention to detail

Certifications: Google Advanced Data Analytics (Coursera) | Google Business Intelligence (Coursera)
MLOps (DeepLearning.AI, Coursera)